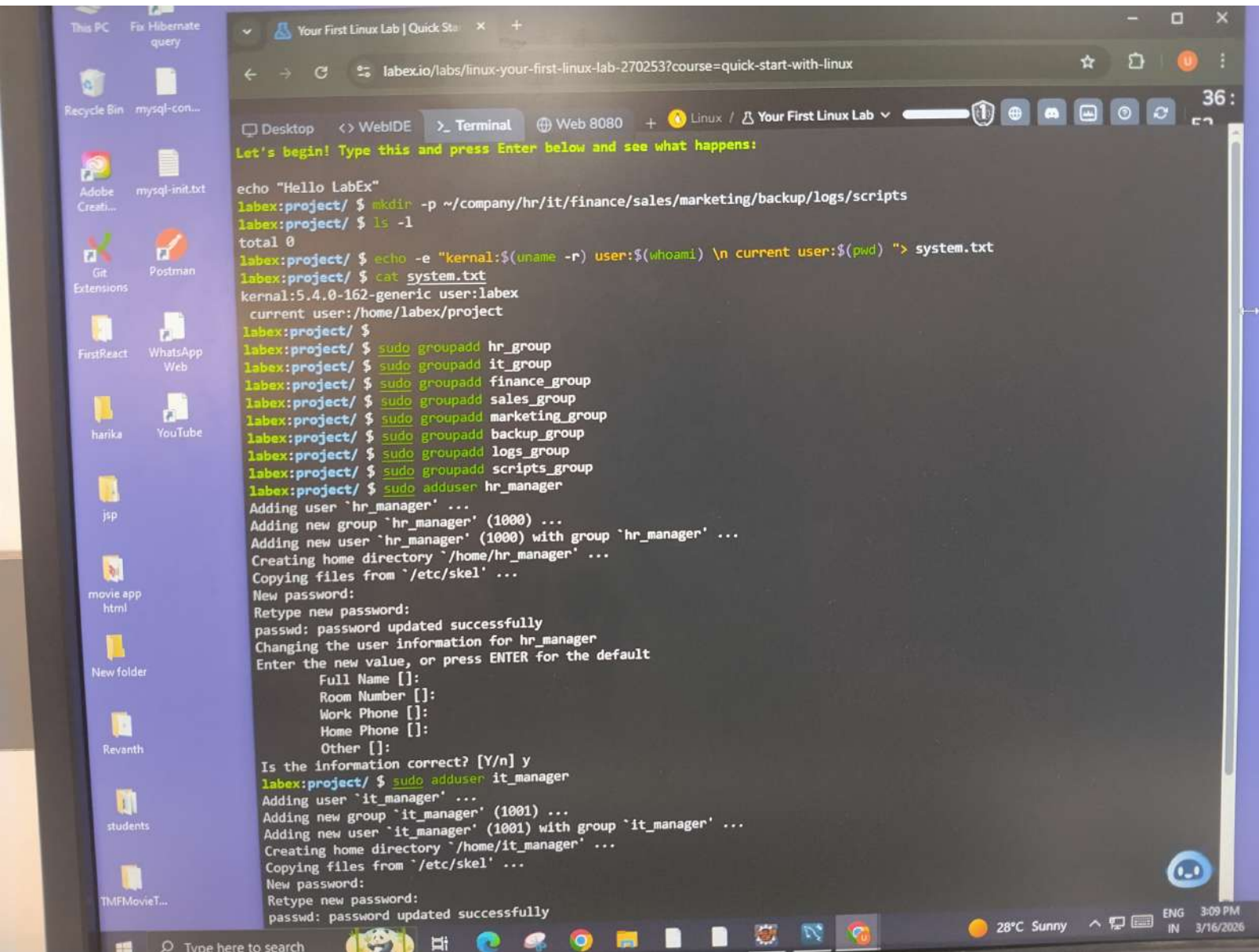




Let's begin! Type this and press Enter below and see what happens:

```
echo "Hello LabEx"
labex:project/ $ mkdir -p ~/company/hr/it/finance/sales/marketing/backup/logs/scripts
labex:project/ $ ls -l
total 0
labex:project/ $ echo -e "kernel:$(uname -r) user:$(whoami) \n current user:$(pwd)" > system.txt
labex:project/ $ cat system.txt
kernel:5.4.0-162-generic user:labex
current user:~/home/labex/project
labex:project/ $
labex:project/ $ sudo groupadd hr_group
labex:project/ $ sudo groupadd it_group
labex:project/ $ sudo groupadd finance_group
labex:project/ $ sudo groupadd sales_group
labex:project/ $ sudo groupadd marketing_group
labex:project/ $ sudo groupadd backup_group
labex:project/ $ sudo groupadd logs_group
labex:project/ $ sudo groupadd scripts_group
labex:project/ $ sudo adduser hr_manager
Adding user `hr_manager' ...
Adding new group `hr_manager' (1000) ...
Adding new user `hr_manager' (1000) with group `hr_manager' ...
Creating home directory `~/home/hr_manager' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for hr_manager
Enter the new value, or press ENTER for the default
Full Name []:
Room Number []:
Work Phone []:
Home Phone []:
Other []:
Is the information correct? [Y/n] y
labex:project/ $ sudo adduser it_manager
Adding user `it_manager' ...
Adding new group `it_manager' (1001) ...
Adding new user `it_manager' (1001) with group `it_manager' ...
Creating home directory `~/home/it_manager' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
```

```
labex:project/ $ cat system.txt
kernel:5.4.0-162-generic user:labex
current user:/home/labex/project
labex:project/ $
labex:project/ $ sudo groupadd hr_group
labex:project/ $ sudo groupadd it_group
labex:project/ $ sudo groupadd finance_group
labex:project/ $ sudo groupadd sales_group
labex:project/ $ sudo groupadd marketing_group
labex:project/ $ sudo groupadd backup_group
labex:project/ $ sudo groupadd logs_group
labex:project/ $ sudo groupadd scripts_group
labex:project/ $ sudo adduser hr_manager
Adding user `hr_manager' ...
Adding new group `hr_manager' (1000) ...
Adding new user `hr_manager' (1000) with group `hr_manager' ...
Creating home directory `/home/hr_manager' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for hr_manager
Enter the new value, or press ENTER for the default
    Full Name []:
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y
labex:project/ $ sudo adduser it_manager
Adding user `it_manager' ...
Adding new group `it_manager' (1001) ...
Adding new user `it_manager' (1001) with group `it_manager' ...
Creating home directory `/home/it_manager' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for it_manager
Enter the new value, or press ENTER for the default
    Full Name []:
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y
labex:project/ $ sudo adduser hr_manager
```

query

Your First Linux Lab | Quick Start

labex.io/labs/linux-your-first-linux-lab-270253?course=quick-start-with-linux

Desktop WebIDE Terminal Web 8080 Linux / Your First Linux Lab

Retype new password:

passwd: password updated successfully

Changing the user information for finance_manager

Enter the new value, or press ENTER for the default

Full Name []:

Room Number []:

Work Phone []:

Home Phone []:

Other []:

Is the information correct? [Y/n] y

labex:project/ \$ sudo adduser sales_manager

Adding user `sales_manager' ...

Adding new group `sales_manager' (1003) ...

Adding new user `sales\_manager' (1003) with group `sales_manager' ...

Creating home directory `/home/sales_manager' ...

Copying files from `/etc/skel' ...

New password:

Retype new password:

passwd: password updated successfully

Changing the user information for sales_manager

Enter the new value, or press ENTER for the default

Full Name []:

Room Number []:

Work Phone []:

Home Phone []:

Other []:

Is the information correct? [Y/n] y

labex:project/ \$ sudo adduser marketing_manager

Adding user `marketing_manager' ...

Adding new group `marketing_manager' (1004) ...

Adding new user `marketing\_manager' (1004) with group `marketing_manager' ...

Creating home directory `/home/marketing_manager' ...

Copying files from `/etc/skel' ...

New password:

Retype new password:

passwd: password updated successfully

Changing the user information for marketing_manager

Enter the new value, or press ENTER for the default

Full Name []:

Room Number []:

Work Phone []:

Home Phone []:

Other []:

Is the information correct? [Y/n] y

labex:project/ \$

Type here to search

DELL

```
Retype new password:
passwd: password updated successfully
Changing the user information for hr_emp1
Enter the new value, or press ENTER for the default
    Full Name []:
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y
labex:project/ $ sudo adduser it_emp1
Adding user `it_emp1' ...
Adding new group `it_emp1' (1006) ...
Adding new user `it_emp1' (1006) with group `it_emp1' ...
Creating home directory `/home/it_emp1' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for it_emp1
Enter the new value, or press ENTER for the default
    Full Name []:
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y
labex:project/ $ sudo adduser sales_emp1
Adding user `sales_emp1' ...
Adding new group `sales_emp1' (1007) ...
Adding new user `sales_emp1' (1007) with group `sales_emp1' ...
Creating home directory `/home/sales_emp1' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for sales_emp1
Enter the new value, or press ENTER for the default
    Full Name []:
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y
labex:project/ $
```

Room Number []:
Work Phone []:
Home Phone []:
Other []:

Is the information correct? [Y/n] y

labex:project/ \$ groups

labex sudo ssl-cert public

labex:project/ \$ sudo usermod -aG hr_group hr_manager

labex:project/ \$ sudo usermod -aG it_group it_manager

labex:project/ \$ sudo usermod -aG finance_group finance_manager

labex:project/ \$ sudo usermod -aG sales_group sales_group

usermod: user "sales_group" does not exist

labex:project/ \$ sudo usermod -aG marketing_group marketing_group

usermod: user "marketing_group" does not exist

labex:project/ \$ sudo usermod -aG sales_group sales_manager

labex:project/ \$ sudo usermod -aG marketing_group marketing_manager

labex:project/ \$ company="tech solutions"

labex:project/ \$ mkdir hr

labex:project/ \$ mkdir it

labex:project/ \$ mkdir finance

labex:project/ \$ mkdir sales

labex:project/ \$ mkdir marketing

labex:project/ \$ touch hr/welcome.txt

labex:project/ \$ touch it/welcome.txt

labex:project/ \$ touch finance/welcome.txt

labex:project/ \$ touch sales/welcome.txt

labex:project/ \$ touch marketing/welcome.txt

labex:project/ \$ sudo chown it_m:it_it

chown: missing operand after "it_m:it_it"

Try "chown --help" for more information.

labex:project/ \$

labex:project/ \$ sudo chown it_manager it_group it

chown: cannot access "it_group": No such file or directory

labex:project/ \$ sudo chown it_manager:it_group it

labex:project/ \$ sudo chown hr_manager:hr_group hr

labex:project/ \$ sudo chown finance_manager:finance_group finance

labex:project/ \$ sudo chown marketing_manager:marketing_group marketing

labex:project/ \$ sudo chown sales_manager:sales_group sales

labex:project/ \$ chmod u+rw,g+wx,o-r hr/welcome.txt

labex:project/ \$ chmod u+rw,g+wx,o-r it/welcome.txt

labex:project/ \$ chmod u+rw,g+wx,o-r sales/welcome.txt

labex:project/ \$ chmod u+rw,g+wx,o-r finance/welcome.txt

labex:project/ \$ chmod u+rw,g+wx,o-r marketing/welcome.txt

labex:project/ \$ nano backup.sh

labex:project/ \$

```
labex:project/ $ sudo chown it_m:it_it
chown: missing operand after 'it_m:it_it'
Try 'chown --help' for more information.
labex:project/ $
labex:project/ $ sudo chown it_manager it_group it
chown: cannot access 'it_group': No such file or directory
labex:project/ $ sudo chown it_manager:it_group it
labex:project/ $ sudo chown hr_manager:hr_group hr
labex:project/ $ sudo chown finance_manager:finance_group fi
labex:project/ $ sudo chown marketing_manager:marketing_grou
labex:project/ $ sudo chown sales_manager:sales_group sales
labex:project/ $ chmod u+rwx,g+wx,o-r hr/welcome.txt
labex:project/ $ chmod u+rwx,g+wx,o-r it/welcome.txt
labex:project/ $ chmod u+rwx,g+wx,o-r sales/welcome.txt
labex:project/ $ chmod u+rwx,g+wx,o-r finance/welcome.txt
labex:project/ $ chmod u+rwx,g+wx,o-r marketing/welcome.txt
labex:project/ $ nano backup.sh
labex:project/ $ tar -czvf document.tar.gz hr
hr/
hr/welcome.txt
labex:project/ $ tar -czvf document.tar.gz it
it/
it/welcome.txt
labex:project/ $ tar -czvf document.tar.gz sales
sales/
sales/welcome.txt
labex:project/ $ tar -czvf document.tar.gz finance
finance/
finance/welcome.txt
labex:project/ $ tar -czvf document.tar.gz marketing
marketing/
marketing/welcome.txt
labex:project/ $
```

here to search



DELL

```

labex:project/ $ free -h
               total        used        free      shared  buff/cache   available
mem:           1561         5.60i       3.00i        91Mi        6.70i        9.30i
labex:project/ $ ss -tuln
State      Recv-Q   Send-Q   Local Address:Port   Peer Address:Port   Process
tcp        UNCONN   0        0      127.0.0.11:49291     0.0.0.0:*
tcp        UNCONN   0        0      0.0.0.0:3001        0.0.0.0:*
tcp        LISTEN   0        128     0.0.0.0:22         0.0.0.0:*
tcp        LISTEN   0        5       0.0.0.0:3001        0.0.0.0:*
tcp        LISTEN   0        128     0.0.0.0:3002        0.0.0.0:*
tcp        LISTEN   0       4096    127.0.0.11:46731    0.0.0.0:*
tcp        LISTEN   0        128     [::]:22            [::]:22
labex:project/ $ du -sh ~/company_server
.8K    /home/labex/company_server
labex:project/ $ ip a
lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
37652: eth0@if837653: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether 02:42:ac:3d:02:04 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 172.61.2.4/16 brd 172.61.255.255 scope global eth0
        valid_lft forever preferred_lft forever
labex:project/ $ ip addr
lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
37652: eth0@if837653: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether 02:42:ac:3d:02:04 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 172.61.2.4/16 brd 172.61.255.255 scope global eth0
        valid_lft forever preferred_lft forever
labex:project/ $ ping -c 3 google.com
PING google.com (142.250.189.206) 56(84) bytes of data.
-- google.com ping statistics ---

```



```
labex:project/ $ df -h
Filesystem      Size  Used Avail Use% Mounted on
overlay         20G  126M   20G   1% /
tmpfs           64M    0   64M   0% /dev
tmpfs           7.7G    0   7.7G   0% /sys/fs/cgroup
shm            64M    0   64M   0% /dev/shm
dev/nvme1n1    100G   20G   81G  20% /etc/hosts
tmpfs          7.7G    0   7.7G   0% /proc/acpi
tmpfs          7.7G    0   7.7G   0% /proc/scsi
tmpfs          7.7G    0   7.7G   0% /sys/firmware
```

```
labex:project/ $ free -h
              total        used        free      shared  buff/cache   available
Mem:          15Gi         5.4Gi         2.7Gi         102Mi         7.3Gi         9.6Gi
Swap:          0B           0B           0B
```

```
labex:project/ $ ss -tuln
State      Recv-Q   Send-Q   Local Address:Port   Peer Address:Port   Process
UNCONN     0         0         127.0.0.11:40735     0.0.0.0:*
UNCONN     0         0         0.0.0.0:3001        0.0.0.0:*
LISTEN     0         4096      127.0.0.11:37909     0.0.0.0:*
LISTEN     0         128       0.0.0.0:22          0.0.0.0:*
LISTEN     0         5         0.0.0.0:3001        0.0.0.0:*
LISTEN     0         128       0.0.0.0:3002        0.0.0.0:*
LISTEN     0         128       [::]:22             [::]:*
```

```
labex:project/ $ ps -eo pid,ppid,cmd,%mem,%cpu --sort=-%cpu | head -6
```

PID	PPID	CMD	%MEM	%CPU
36	1	/usr/bin/Xvnc :1 -disableBa	0.6	0.4
368	367	zsh -i	0.0	0.1
1	0	/bin/bash /etc/shiyanlou/sb	0.0	0.0
21	1	/usr/bin/python3 /usr/bin/s	0.1	0.0
22	21	sshd: /usr/sbin/sshd -D [li	0.0	0.0

```
labex:project/ $ uptime
22:28:32 up 194 days,  5:49,  0 users,  load average: 0.69, 0.86, 0.73
```

```
labex:project/ $
```

```
labex:project/ $ mkdir ~/my_website
labex:project/ $ mkdir ~/my_website/{html,backups,logs,scripts,posts}
labex:project/ $ touch ~/my_website/server_info.txt
labex:project/ $ hostname >> ~/my_website/server_info.txt
labex:project/ $ uname -r >> ~/my_website/server_info.txt

labex:project/ $ date >> ~/my_website/server_info.txt
labex:project/ $ sudo apt install apache2

Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
apache2 is already the newest version (2.4.52-1ubuntu4.11).
The following packages were automatically installed and are no longer required:
  gir1.2-libxfce4util-1.0 gir1.2-xfconf-0 gsfonts gsfonts-x11 libdbus-glib-1-2 libgdk-pixbuf-xlib-2.0-0 libjpeg-turbo-progs miscfiles xscreensaver-data
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
labex:project/ $ sudo systemctl status apache2
System has not been booted with systemd as init system (PID 1). Can't operate.
Failed to connect to bus: Host is down
labex:project/ $ sudo systemctl enable apache2
Synchronizing state of apache2.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable apache2
labex:project/ $ nano ~/my_website/html/index.html
labex:project/ $ nano ~/my_website/html/about.html
labex:project/ $ sudo cp ~/my_website/html/index.html /var/www/html/
labex:project/ $ sudo cp ~/my_website/html/about.html /var/www/html/
labex:project/ $ chmod +x ~/my_website/scripts/new_post.sh
chmod: cannot access '/home/labex/my_website/scripts/new_post.sh': No such file or directory
labex:project/ $
```

GNU nano 6.2

```
<!DOCTYPE html>
<html>
<head>
  <title>About Me</title>
</head>
<body>

  <h1>About Me</h1>

  <p>Name: Ushasree</p>
  <p>I am a diploma student.</p>
  <p>I am learning Linux and basic computer concepts.</p>
  <p>I am interested in technology and improving my skills.</p>

</body>
</html>
```

GNU nano 0.2

/home/taoex/my_w

```
!DOCTYPE html>
```

```
html>
```

```
head>
```

```
  <title>My Linux Blog</title>
```

```
/head>
```

```
body>
```

```
  <h1>Ushasree</h1>
```

```
  <h2>My Linux Blog</h2>
```

```
  <p>Date: 18 March 2026</p>
```

```
  <h3>Blog Posts</h3>
```

```
  <ul>
```

```
    <li>Introduction to Linux</li>
```

```
    <li>Basic Linux Commands</li>
```

```
  </ul>
```

```
/body>
```

```
/html>
```

Ready to start your first linux lab?

Let's begin! Type this and press Enter below and see what happens:

echo "Hello LabEx"

```
labex:project/ $ ~/ my_website
zsh: permission denied: /home/labex/
labex:project/ $ mkdir ~/my_website
labex:project/ $ mkdir ~/my_website/{html, backups,logs,scripts,posts}
labex:project/ $ mkdir ~/my_website/{html,backups,logs,scripts,posts}
labex:project/ $ touch ~/my_website/server_info.txt
labex:project/ $ hostname >> ~/my_website/server_info.txt
labex:project/ $ uname -r >> ~/my_website/server_info.txt
labex:project/ $ date >> ~/my_website/server_info.txt
labex:project/ $ sudo apt install apache2
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
apache2 is already the newest version (2.4.52-1ubuntu4.11).
The following packages were automatically installed and are no longer required:
  gir1.2-libxfce4util-1.0 gir1.2-xfconf-0 gsfonts gsfonts-x11 libdbus-glib-1-2 libgdk-pixbuf-xlib-2.0-0 libjpeg-turbo-progs miscfiles xscreensaver-data
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
labex:project/ $ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
839412: eth0@if839413: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether 02:42:ac:3d:01:43 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 172.61.1.67/16 brd 172.61.255.255 scope global eth0
        valid_lft forever preferred_lft forever
labex:project/ $
```

```

touch: cannot touch '/home/labex/my_website/server_info.txt': No such file or directory
labex:project/ $ touch ~/my_website/server_info.txt
touch: cannot touch '/home/labex/my_website/server_info.txt': No such file or directory
labex:project/ $ pwd
/home/labex/project
labex:project/ $ cd ~
labex:~/ $ pwd
/home/labex
labex:~/ $ mkdir my_website
labex:~/ $ cd my_website
labex:my_website/ $ touch server_info.txt
labex:my_website/ $ ls
server_info.txt
labex:my_website/ $ echo -e "hostname: $(hostname) \n
dquote>
dquote>
labex:my_website/ $ echo -e "hostname: $(hostname) \nkernal: $(uname -r) \ndate: $(date)" > server_info.txt

labex:my_website/ $ cat server_info.txt
hostname: 69ba7bc9a829da522a2a6e9c
kernal: 5.4.0-162-generic
date: Wed Mar 18 18:29:28 CST 2026
labex:my_website/ $ sudo apt update
Get:1 http://mirrors.cloud.aliyuncs.com/ubuntu jammy InRelease [270 kB]
Get:2 http://mirrors.cloud.aliyuncs.com/ubuntu jammy-updates InRelease [128 kB]
Get:3 http://mirrors.cloud.aliyuncs.com/ubuntu jammy-backports InRelease [127 kB]
Get:4 http://mirrors.cloud.aliyuncs.com/ubuntu jammy-security InRelease [129 kB]
Get:5 http://mirrors.cloud.aliyuncs.com/ubuntu jammy/universe amd64 Packages [17.5 MB]
Get:6 http://mirrors.cloud.aliyuncs.com/ubuntu jammy/restricted amd64 Packages [164 kB]
Get:7 http://mirrors.cloud.aliyuncs.com/ubuntu jammy/multiverse amd64 Packages [266 kB]
Get:8 http://mirrors.cloud.aliyuncs.com/ubuntu jammy/main amd64 Packages [1792 kB]
Get:9 http://mirrors.cloud.aliyuncs.com/ubuntu jammy-updates/restricted amd64 Packages [7002 kB]
Get:10 http://mirrors.cloud.aliyuncs.com/ubuntu jammy-updates/universe amd64 Packages [1615 kB]
Get:11 http://mirrors.cloud.aliyuncs.com/ubuntu jammy-updates/multiverse amd64 Packages [70.9 kB]
Get:12 http://mirrors.cloud.aliyuncs.com/ubuntu jammy-updates/main amd64 Packages [4164 kB]
Get:13 http://mirrors.cloud.aliyuncs.com/ubuntu jammy-backports/universe amd64 Packages [35.6 kB]
Get:14 http://mirrors.cloud.aliyuncs.com/ubuntu jammy-backports/main amd64 Packages [84.0 kB]
Get:15 http://mirrors.cloud.aliyuncs.com/ubuntu jammy-security/universe amd64 Packages [1303 kB]
Get:16 http://mirrors.cloud.aliyuncs.com/ubuntu jammy-security/main amd64 Packages [3830 kB]

```

```
Get:8 http://mirrors.cloud.aliyuncs.com/ubuntu jammy/main amd64 Packages [1152 kB]
Get:9 http://mirrors.cloud.aliyuncs.com/ubuntu jammy-updates/restricted amd64 Packages [7002 kB]
Get:10 http://mirrors.cloud.aliyuncs.com/ubuntu jammy-updates/universe amd64 Packages [1615 kB]
Get:11 http://mirrors.cloud.aliyuncs.com/ubuntu jammy-updates/multiverse amd64 Packages [70.9 kB]
Get:12 http://mirrors.cloud.aliyuncs.com/ubuntu jammy-updates/main amd64 Packages [4164 kB]
Get:13 http://mirrors.cloud.aliyuncs.com/ubuntu jammy-backports/universe amd64 Packages [35.6 kB]
Get:14 http://mirrors.cloud.aliyuncs.com/ubuntu jammy-backports/main amd64 Packages [84.0 kB]
Get:15 http://mirrors.cloud.aliyuncs.com/ubuntu jammy-security/universe amd64 Packages [1303 kB]
Get:16 http://mirrors.cloud.aliyuncs.com/ubuntu jammy-security/main amd64 Packages [3830 kB]
Get:17 http://mirrors.cloud.aliyuncs.com/ubuntu jammy-security/restricted amd64 Packages [6786 kB]
Get:18 http://mirrors.cloud.aliyuncs.com/ubuntu jammy-security/multiverse amd64 Packages [62.6 kB]
Fetched 45.3 MB in 3s (15.0 MB/s)
```

```
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed by your system:
gir1.2-libxfce4util-1.0 gir1.2-xfconf-0 gsf
Use 'sudo apt autoremove' to remove them.
```

```
Desktop  <> WebIDE  Terminal  Web 8080  +  Linux 7  Your First Linux Lab 7

Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
338 packages can be upgraded. Run 'apt list --upgradable' to see them.
labex:my_website/ $ sudo apt install apache2
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  gir1.2-libxfce4util-1.0 gir1.2-xfconf-0 gsfonts gsfonts-x11 libdbus-glib-1-2 libgdk-pixbuf-xlib-2.0-0 libjpeg-turbo-progs miscfiles xscreensaver-data
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom ufw
The following packages will be upgraded:
  apache2 apache2-bin apache2-data apache2-utils
4 upgraded, 0 newly installed, 0 to remove and 334 not upgraded.
Need to get 1712 kB of archives.
After this operation, 31.7 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://mirrors.cloud.aliyuncs.com/ubuntu jammy-updates/main amd64 apache2 amd64 2.4.52-1ubuntu4.19 [97.9 kB]
Get:2 http://mirrors.cloud.aliyuncs.com/ubuntu jammy-updates/main amd64 apache2-bin amd64 2.4.52-1ubuntu4.19 [1360 kB]
Get:3 http://mirrors.cloud.aliyuncs.com/ubuntu jammy-updates/main amd64 apache2-data all 2.4.52-1ubuntu4.19 [165 kB]
Get:4 http://mirrors.cloud.aliyuncs.com/ubuntu jammy-updates/main amd64 apache2-utils amd64 2.4.52-1ubuntu4.19 [89.5 kB]
Fetched 1712 kB in 0s (37.6 MB/s)
(Reading database ... 147587 files and directories currently installed.)
Preparing to unpack .../apache2_2.4.52-1ubuntu4.19_amd64.deb ...
Unpacking apache2 (2.4.52-1ubuntu4.19) over (2.4.52-1ubuntu4.11) ...
Preparing to unpack .../apache2-bin_2.4.52-1ubuntu4.19_amd64.deb ...
Unpacking apache2-bin (2.4.52-1ubuntu4.19) over (2.4.52-1ubuntu4.11) ...
Preparing to unpack .../apache2-data_2.4.52-1ubuntu4.19_all.deb ...
Unpacking apache2-data (2.4.52-1ubuntu4.19) over (2.4.52-1ubuntu4.11) ...
Preparing to unpack .../apache2-utils_2.4.52-1ubuntu4.19_amd64.deb ...
Unpacking apache2-utils (2.4.52-1ubuntu4.19) over (2.4.52-1ubuntu4.11) ...
Setting up apache2-bin (2.4.52-1ubuntu4.19) ...
Setting up apache2-data (2.4.52-1ubuntu4.19) ...
Setting up apache2-utils (2.4.52-1ubuntu4.19) ...
Setting up apache2 (2.4.52-1ubuntu4.19) ...
```

```
labex:my_website/ $ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
787472: eth0@if787473: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether 02:42:ac:3d:01:72 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 172.61.1.114/16 brd 172.61.255.255 scope global eth0
        valid_lft forever preferred_lft forever
```

Desktop

WebIDE

Terminal

GNU nano 6.2

#!/bin/bash

echo "Enter post title:"

read title

echo "Your post title is: \$title"

```

labex:my_website/ $ touch index.html
labex:my_website/ $ touch counter.sh
labex:my_website/ $ nano counter.sh
labex:my_website/ $ echo 0 > post/visitors.txt
labex:my_website/ $ cat post/visitors.txt
0
labex:my_website/ $ count=$(cat post/visitors.txt)
count=$((count+1))
echo $count > post/visitors.txt
labex:my_website/ $ nano counter.sh
labex:my_website/ $ chmod +x counter.sh
labex:my_website/ $ ./counter.sh
Visitor count updated: 1
labex:my_website/ $ ./counter.sh
Visitor count updated: 2
labex:my_website/ $ ./counter.sh
Visitor count updated: 3
labex:my_website/ $ ./counter.sh
Visitor count updated: 4
labex:my_website/ $ echo "Visit: $count | Date: $(date) | IP: $(hostname -I)" >> visits.log
labex:my_website/ $ ./counter.sh
Visitor count updated: 5
labex:my_website/ $ nano counter.sh
labex:my_website/ $ ./counter.sh
Visitor count updated: 6
labex:my_website/ $ count=$(cat post/visitors.
count=$((count+1))
echo $count > post/visitors.txt
labex:my_website/ $ cat visits.log
Visit: 1 | Date: Wed Mar 18 18:54:59 CST 2026 | IP: 172.61.1.114
Visit: 2 | Date: Wed Mar 18 18:55:25 CST 2026 | IP: 172.61.1.114
Visit: 3 | Date: Wed Mar 18 18:55:30 CST 2026 | IP: 172.61.1.114
Visit: 4 | Date: Wed Mar 18 18:55:34 CST 2026 | IP: 172.61.1.114
Visit: 1 | Date: Wed Mar 18 18:58:03 CST 2026 | IP: 172.61.1.114
Visit: 5 | Date: Wed Mar 18 18:58:17 CST 2026 | IP: 172.61.1.114
Visit: 6 | Date: Wed Mar 18 19:00:23 CST 2026 | IP: 172.61.1.114
Visit: 6 | Date: Wed Mar 18 19:00:23 CST 2026 | IP: 172.61.1.114
labex:my_website/ $ touch index.html

```

```
Desktop  WebIDE  Terminal  Web 8080  Linux / Your First Linux Lab
echo $count > post/visitors.txt
labex:my_website/ $ nano counter.sh
labex:my_website/ $ chmod +x counter.sh
labex:my_website/ $ ./counter.sh
Visitor count updated: 1
labex:my_website/ $ ./counter.sh
Visitor count updated: 2
labex:my_website/ $ echo "Visit: $count | Date: $(date) | IP: $(hostname -I)" >> visits.log
Visit: 3 | Date: Wed Mar 18 18:55:30 CST 2026 | IP: 172.61.1.114
Visit: 4 | Date: Wed Mar 18 18:55:34 CST 2026 | IP: 172.61.1.114
Visit: 1 | Date: Wed Mar 18 18:58:03 CST 2026 | IP: 172.61.1.114
Visit: 5 | Date: Wed Mar 18 18:58:17 CST 2026 | IP: 172.61.1.114
Visit: 6 | Date: Wed Mar 18 19:00:23 CST 2026 | IP: 172.61.1.114
Visit: 6 | Date: Wed Mar 18 19:00:23 CST 2026 | IP: 172.61.1.114
labex:my_website/ $ nano backup_website.sh
labex:my_website/ $ chmod +x backup_website.sh
labex:my_website/ $ ./backup_website.sh
tar: Removing leading '/' from member names
tar: posts: Cannot stat: No such file or directory
tar: Exiting with failure status due to previous errors
Backups created:
apache_backup_2026-03-18_19-04-12.tar.gz  project_backup_2026-03-18_19-04-12.tar.gz
labex:my_website/ $ mkdir posts
labex:my_website/ $ ./backup_website.sh
tar: Removing leading '/' from member names
Backups created:
apache_backup_2026-03-18_19-04-12.tar.gz  apache_backup_2026-03-18_19-05-59.tar.gz  project_backup_2026-03-18_19-04-12.tar.gz  project_backup_2026-03-18_19-05-59.tar.gz
labex:my_website/ $ nano monitor_website.sh
labex:my_website/ $ chmod +x monitor_website.sh
labex:my_website/ $ ./monitor_website.sh
System has not been booted with systemd as init system (PID 1). Can't operate.
Failed to connect to bus: Host is down
System has not been booted with systemd as init system (PID 1). Can't operate.
Failed to connect to bus: Host is down
labex:my_website/ $ cat monitor.log
----- Wed Mar 18 19:07:43 CST 2026 -----
Apache is NOT running. Restarting...
Disk Usage:
Filesystem      Size  Used Avail Use% Mounted on
```

```
labex:my_website/ $ cat monitor.log
```

```
----- Wed Mar 18 19:07:43 CST 2026 -----
```

```
Apache is NOT running. Restarting...
```

```
Disk Usage:
```

Filesystem	Size	Used	Avail	Use%	Mounted on
overlay	20G	213M	20G	2%	/

```
Memory Usage:
```

	total	used	free	shared	buff/cache	available
Mem:	15Gi	5.2Gi	2.0Gi	89Mi	8.2Gi	9.7Gi
Swap:	0B	0B	0B			

```
Website Status:
```

```
-----
```